

Foundations of Probability

A collection of definitions, theorems, and formulae

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1 Random variables

1.1 Discrete distributions

Let Ω be finite or countably infinite, say $\Omega = \{\omega_0, \omega_1, \dots\}$ and define a discrete random variable as a function $x : \Omega \rightarrow \mathcal{X}$ which then has the support values x_j and probability mass function

$$x(\omega_j) = x_j, \quad p_x(x_j) = \mathbb{P}(x = x_j), \quad j = 0, 1, \dots,$$

where

$$0 \leq p_x(x_j) \leq 1, \quad \sum_j p_x(x_j) = 1.$$

Given a deterministic function f of x , its expectation is

$$\mathbb{E}_{p_x}[f(x)] = \sum_j f(x_j) p_x(x_j).$$

The r -th raw moment of the distribution is

$$m_r = \mathbb{E}_{p_x}[x^r] = \sum_j x_j^r p_x(x_j).$$

These raw moments are generated by the MGF

$$M_x(t) = \mathbb{E}_{p_x}[e^{tx}] = \sum_j e^{tx_j} p_x(x_j),$$

whenever the sum is finite. In particular, if $M_x(t)$ is finite in a neighborhood of $t = 0$, then we can use these to generate the moments m_r using

$$m_r = M_x^{(r)}(0), \quad m_1 = M'_x(0), \quad m_2 = M''_x(0).$$

The variance of x is

$$\text{Var}_{p_x}(x) = m_2 - m_1^2 = M''_x(0) - (M'_x(0))^2 = \sum_j (x_j - m_1)^2 p_x(x_j),$$

where $m_r = \mathbb{E}_{p_x}[x^r]$.

1.2 Continuous random variables

Let Ω be a continuous space, say $\Omega = \mathbb{R}$ and define a continuous random variable as $x : \Omega \rightarrow \mathcal{X}$, then its distribution is described by a density $p_x : \mathcal{X} \rightarrow \mathbb{R}$, where

$$p_x(x) \geq 0, \quad \int_{\mathcal{X}} p_x(x) dx = 1.$$

For any interval $A \subseteq \mathcal{X}$,

$$\mathbb{P}(x \in A) = \int_{x \in A} p_x(x) dx.$$

Given a deterministic function f of x , its expectation is

$$\mathbb{E}_{p_x}[f(x)] = \int_{\mathcal{X}} f(x) p_x(x) dx.$$

The r -th raw moment of the distribution is

$$m_r = \mathbb{E}_{p_x}[x^r] = \int_{\mathcal{X}} x^r p_x(x) dx.$$

These raw moments are generated by the moment-generating function

$$M_x(t) = \mathbb{E}_{p_x}[e^{tx}] = \int_{\mathcal{X}} e^{tx} p_x(x) dx,$$

whenever the integral is finite. In particular, if $M_x(t)$ is finite in a neighborhood of $t = 0$, then

$$m_r = M_x^{(r)}(0), \quad m_1 = M'_x(0), \quad m_2 = M''_x(0).$$

The variance of x is

$$\text{Var}_{p_x}(x) = m_2 - m_1^2 = \int_{\mathcal{X}} (x - m_1)^2 p_x(x) dx,$$

where $m_r = \mathbb{E}_{p_x}[x^r]$.

1.3 Normal distribution

If $x \sim \mathcal{N}(\mu, \sigma^2)$, then the density of x is

$$p_x(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right).$$

The CDF of x is

$$F_x(x) = \mathbb{P}(x \leq x) = \int_{-\infty}^x \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(s - \mu)^2}{2\sigma^2}\right) ds.$$

For the standard normal random variable $z \sim \mathcal{N}(0, 1)$, define

$$\Phi(z) = \mathbb{P}(z \leq z).$$

Then

$$\mathbb{P}(x \leq x) = \Phi\left(\frac{x - \mu}{\sigma}\right).$$

Properties of the standard normal CDF:

- Limits: $\lim_{z \rightarrow \infty} \Phi(z) = 1$ and $\lim_{z \rightarrow -\infty} \Phi(z) = 0$.
- Symmetry: $\Phi(-z) = 1 - \Phi(z)$.
- Standardization: if $x \sim \mathcal{N}(\mu, \sigma^2)$, then $\mathbb{P}(x \leq x) = \Phi((x - \mu)/\sigma)$.
- Φ is not elementary in closed form; it is expressed using the error function erf.
- Φ is increasing, so Φ^{-1} is also increasing.

The MGF is

$$M_x(t) = \mathbb{E}_{p_x}[e^{tx}] = \exp\left(\mu t + \frac{1}{2}\sigma^2 t^2\right).$$

1.4 Bernoulli distribution

The Bernoulli distribution is defined by

$$p_x(j) = \mathbb{P}(x = j) = \begin{cases} p, & j = 1, \\ q, & j = 0, \end{cases}$$

where $p + q = 1$ and $p, q \geq 0$. Then

$$\mathbb{E}_{p_x}[x] = p, \quad \text{Var}_{p_x}(x) = pq.$$

The MGF is

$$M_x(t) = \mathbb{E}_{p_x}[e^{tx}] = q + pe^t.$$

1.5 Binomial distribution

A discrete random variable x follows a binomial distribution with parameters $n \in \mathbb{N}$ and $p \in [0, 1]$, written

$$x \sim \text{Binomial}(n, p),$$

if x represents the number of successes in n independent Bernoulli trials, each with success probability p . The PMF is

$$p_x(k) = \mathbb{P}(x = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \dots, n.$$

Writing $q = 1 - p$, we have

$$\mathbb{E}_{p_x}[x] = np, \quad \text{Var}_{p_x}(x) = npq.$$

The moment generating function is

$$M_x(t) = \mathbb{E}_{p_x}[e^{tx}] = (q + pe^t)^n.$$

Remarks:

- Models the count of successes in a fixed number of independent trials.
- As $n \rightarrow \infty$ and $p \rightarrow 0$ with $\lambda = np$ fixed, the binomial distribution converges to $\text{Poisson}(\lambda)$.
- For large n , by the Central Limit Theorem, x is approximately normal with mean np and variance npq .

1.6 Multinomial distribution

The multinomial distribution generalizes the binomial distribution over k categories. In particular, if we have a k -sided die rolled n times, then a multinomial distribution models the probability that any particular combination of rolls across the k categories appears.

Let

$$x = (x_1, \dots, x_k) \in \mathbb{N}_0^k, \quad x \sim \text{Multinomial}(n, \mathbf{p}),$$

where the deterministic parameter is

$$\mathbf{p} = (p_1, \dots, p_k) \in \mathbb{R}^k, \quad p_i \geq 0, \quad \sum_{i=1}^k p_i = 1.$$

The parameter p_i is interpreted as the probability of category i on each of the n trials. The

PMF is

$$p_{\mathbf{x}}(\mathbf{x}) = \mathbb{P}(x = \mathbf{x}) = \binom{n}{x_1, \dots, x_k} \prod_{i=1}^k p_i^{x_i} = \frac{n!}{\prod_{i=1}^k x_i!} \prod_{i=1}^k p_i^{x_i},$$

where

$$x_i \in \mathbb{N}_0, \quad \sum_{i=1}^k x_i = n.$$

The moments are

$$\mathbb{E}_{p_{\mathbf{x}}}[x_i] = np_i, \quad \text{Var}_{p_{\mathbf{x}}}(x_i) = np_i(1 - p_i), \quad \text{Cov}_{p_{\mathbf{x}}}(x_i, x_j) = -np_i p_j \quad (i \neq j).$$

The covariance matrix is

$$\text{Cov}_{p_{\mathbf{x}}}(x) = n(\text{diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}^\top).$$

The probability generating function and moment generating function are

$$G_{\mathbf{x}}(\mathbf{t}) = \mathbb{E}_{p_{\mathbf{x}}} \left[\prod_{i=1}^k t_i^{x_i} \right] = \left(\sum_{i=1}^k p_i t_i \right)^n,$$

and

$$M_{\mathbf{x}}(\mathbf{t}) = \mathbb{E}_{p_{\mathbf{x}}} \left[e^{\mathbf{t}^\top x} \right] = \left(\sum_{i=1}^k p_i e^{t_i} \right)^n.$$

Common facts:

- Marginally, $x_i \sim \text{Binomial}(n, p_i)$.
- Conditionally on $x_k = x_k$, the remaining counts satisfy

$$x_{-k} \mid x_k = x_k \sim \text{Multinomial} \left(n - x_k, \frac{\mathbf{p}_{-k}}{1 - p_k} \right).$$

1.7 Poisson distribution

The Poisson distribution $\mathcal{P}(\lambda)$ expresses the probability of a given number of events occurring in a fixed interval of time if these events occur with a known constant average rate and independently of the time since the last event. If $x \sim \mathcal{P}(\lambda)$, then

$$p_x(k) = \mathbb{P}(x = k) = \frac{\lambda^k}{k!} e^{-\lambda}, \quad k = 0, 1, \dots$$

where λ is the average rate. Intuition:

- The Poisson distribution counts how many events occur in a Poisson process, while the gamma and exponential distributions describe waiting times between events.
- A Poisson process is a stochastic model used to count random, independent events occurring in continuous time or space at a constant average rate.

The mean and variance are

$$\mathbb{E}_{p_x}[x] = \text{Var}_{p_x}(x) = \lambda.$$

Poisson is the limit of binomial: If $n \rightarrow \infty$, $p \rightarrow 0$, and fix $np = \lambda$ as constant, then

$$\binom{n}{k} p^k q^{n-k} \rightarrow \frac{\lambda^k}{k!} e^{-\lambda}.$$

1.8 Gamma distribution

Let $x \sim \text{Gamma}(\alpha, \lambda)$, where $\alpha > 0$ is the shape and $\lambda > 0$ is the rate. Its PDF is

$$p_x(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}, \quad x \geq 0.$$

- The gamma distribution models positive continuous quantities, especially waiting times.
- If events arrive according to a Poisson process with rate λ , then $\text{Gamma}(\alpha, \lambda)$ models the waiting time until α events when α is an integer.
- The shape α controls how concentrated the distribution is, while the rate λ rescales time: larger λ means shorter waiting times.

Its mean and variance are

$$\mathbb{E}_{p_x}[x] = \frac{\alpha}{\lambda}, \quad \text{Var}_{p_x}(x) = \frac{\alpha}{\lambda^2}.$$

1.9 Exponential distribution

The exponential distribution is the one-event waiting-time case of the gamma distribution. Let $x \sim \text{Exponential}(\lambda)$, where $\lambda > 0$ is the rate.

$$\text{Exponential}(\lambda) = \text{Gamma}(1, \lambda).$$

Indeed, setting $\alpha = 1$ in the gamma PDF gives the exponential PDF

$$p_x(x) = \frac{\lambda^1}{\Gamma(1)} x^0 e^{-\lambda x} = \lambda e^{-\lambda x}, \quad x \geq 0.$$

- The exponential distribution models the waiting time until the first event in a Poisson process with rate λ .

Its mean and variance are

$$\mathbb{E}_{p_x}[x] = \frac{1}{\lambda}, \quad \text{Var}_{p_x}(x) = \frac{1}{\lambda^2}.$$

Its CDF and survival function are

$$\mathbb{P}(x \leq x) = 1 - e^{-\lambda x}, \quad \mathbb{P}(x > x) = e^{-\lambda x}.$$

A key property of the exponential distribution is memorylessness:

$$\mathbb{P}(x > s + t \mid x > s) = \mathbb{P}(x > t) = e^{-\lambda t}.$$

Gamma waiting times add exponential waiting times:

$$x_1, \dots, x_n \stackrel{\text{iid}}{\sim} \text{Exponential}(\lambda) \implies \sum_{i=1}^n x_i \sim \text{Gamma}(n, \lambda).$$

This means the gamma distribution is the distribution of cumulative waiting time, while the exponential distribution is the distribution of a single waiting time.

1.10 Chi-squared distribution

Let $x \sim \chi_k^2$, where $k \in \mathbb{N}$ is the number of degrees of freedom. Its PDF is

$$p_x(x) = \frac{1}{2^{k/2}\Gamma(k/2)} x^{k/2-1} e^{-x/2}, \quad x \geq 0.$$

- The chi-squared distribution models sums of squared standard Gaussian random variables, where the parameter k counts the number of independent squared Gaussian components:

$$\sum_{i=1}^k z_i^2 \sim \chi_k^2 \quad \text{when} \quad z_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1).$$

As k increases, the distribution becomes more concentrated around k .

- The chi-squared distribution is a special case of the gamma distribution:

$$\chi_k^2 = \text{Gamma}\left(\frac{k}{2}, \frac{1}{2}\right).$$

The chi-squared mean and variance are

$$\mathbb{E}_{p_x}[x] = k, \quad \text{Var}_{p_x}(x) = 2k.$$

2 Random variable conjugacy

In broad strokes, random variable conjugacy describes relations and operations that map distributions to distributions; this contrasts with conjugate priors, which describe instances in Bayesian modeling that $\text{prior} \times \text{likelihood} = \text{posterior} = \text{prior}$.

2.1 L2 norm of iid Gaussians $\rightarrow$ chi-squared

Let $\mathbf{z} \in \mathbb{R}^k$ be a centered and normalized k -dimensional Gaussian random vector:

$$\mathbf{z} \sim \mathcal{N}(0, I_k).$$

Then

$$\|\mathbf{z}\|_2^2 = \sum_{i=1}^k z_i^2 \sim \chi_k^2.$$

More generally, if $\mathbf{x} \in \mathbb{R}^k$ is a k -dimensional Gaussian random vector with

$$\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma), \quad \Sigma \succ 0,$$

then the squared Mahalanobis distance

$$d^2 := \left\| \Sigma^{-1/2}(\mathbf{x} - \boldsymbol{\mu}) \right\|_2^2 = (\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})$$

satisfies

$$d^2 \sim \chi_k^2.$$

2.2 Chi-squared $\rightarrow$ gamma

A chi-squared random variable is a special case of a gamma random variable. If

$$x \sim \chi_k^2,$$

then

$$x \sim \text{Gamma}\left(\frac{k}{2}, \frac{1}{2}\right),$$

where the parameters are shape and rate, respectively. In particular, the density is

$$p_x(x) = \frac{(1/2)^{k/2}}{\Gamma(k/2)} x^{k/2-1} e^{-x/2}, \quad x > 0.$$

2.3 Sums of iid chi-squared $\rightarrow$ chi-squared

If

$$x_1 \sim \chi_{k_1}^2, \quad x_2 \sim \chi_{k_2}^2, \quad x_1 \perp\!\!\!\perp x_2,$$

then

$$x_1 + x_2 \sim \chi_{k_1+k_2}^2.$$

More generally, independent chi-squared random variables add by adding their degrees of freedom:

$$\sum_{i=1}^n x_i \sim \chi_{\sum_{i=1}^n k_i}^2.$$

2.4 Gaussian sample variance $\rightarrow$ chi-squared

Let

$$x_1, \dots, x_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2).$$

Define the sample mean and unbiased sample covariance

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \quad s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2.$$

Then a scaled sample covariance is chi-squared:

$$\frac{(n-1)s^2}{\sigma^2} \sim \chi_{n-1}^2.$$

2.5 Exponential $\rightarrow$ gamma

The exponential distribution is a special case of the gamma distribution. If

$$x \sim \text{Exponential}(\lambda),$$

then

$$x \sim \text{Gamma}(1, \lambda),$$

where the second parameter is the rate. More generally, if

$$x_1, \dots, x_n \stackrel{\text{i.i.d.}}{\sim} \text{Exponential}(\lambda),$$

then

$$\sum_{i=1}^n x_i \sim \text{Gamma}(n, \lambda).$$

2.6 Sums of independent gamma variables with same rate $\rightarrow$ gamma

If

$$x \sim \text{Gamma}(\alpha_1, \lambda), \quad y \sim \text{Gamma}(\alpha_2, \lambda), \quad x \perp\!\!\!\perp y,$$

then

$$x + y \sim \text{Gamma}(\alpha_1 + \alpha_2, \lambda).$$

Gamma variables with the same rate add by adding their shape parameters.

2.7 Ratio of independent gammas $\rightarrow$ beta

If

$$x \sim \text{Gamma}(\alpha, \lambda), \quad y \sim \text{Gamma}(\beta, \lambda), \quad x \perp\!\!\!\perp y,$$

then

$$\frac{x}{x+y} \sim \text{Beta}(\alpha, \beta).$$

This is useful because beta random variables can be constructed by normalizing independent gamma random variables.

3 Tools

3.1 Jensen's inequality for random variables

If x is a random variable and φ is convex, then Jensen's inequality says

$$\varphi(\mathbb{E}_{p_x}[x]) \leq \mathbb{E}_{p_x}[\varphi(x)].$$

In words, the function of the expectation is less than the expectation of the function.

3.2 Integration over product spaces

We collect here some basic manipulations of integrals on products of spaces. Start with the integral over $\mathcal{X} \times \mathcal{Y}$:

$$I = \int_{\mathcal{X} \times \mathcal{Y}} f(x, y) \, d(x, y).$$

(1) We can express the integral as an iterated integral (Fubini's theorem):

$$I = \int_{\mathcal{Y}} \left(\int_{\mathcal{X}} f(x, y) \, dx \right) dy = \int_{\mathcal{X}} \left(\int_{\mathcal{Y}} f(x, y) \, dy \right) dx$$

(2) If $\mathcal{Y}$ is finite, replace the integral over $\mathcal{Y}$ with a sum:

$$I = \sum_{y \in \mathcal{Y}} \int_{\mathcal{X}} f(x, y) \, dx.$$

(3) If $\mathcal{Y}$ is finite, we can condition on each value of y :

$$\int_{\mathcal{X}} f(x, y_1) \, dx + \int_{\mathcal{X}} f(x, y_2) \, dx \quad \text{if } \mathcal{Y} = \{y_1, y_2\}.$$

3.3 Expectation under a marginal versus a joint law

Suppose x and y have joint density $p_{x,y}$, and let p_x be the marginal density of x :

$$p_x(x) = \int_{\mathcal{Y}} p_{x,y}(x, y) \, dy.$$

Then for any function f that only depends on x ,

$$\mathbb{E}_{p_{x,y}}[f(x)] = \mathbb{E}_{p_x}[f(x)].$$

Indeed,

$$\mathbb{E}_{p_{x,y}}[f(x)] = \int_{\mathcal{X}} \int_{\mathcal{Y}} f(x) p_{x,y}(x, y) \, dy \, dx = \int_{\mathcal{X}} f(x) p_x(x) \, dx.$$

The key condition is that p_x must be the marginal distribution induced by the joint law $p_{x,y}$.

3.4 Probability as the expectation of an indicator function

For any event $\mathcal{A}$,

$$\mathbb{P}(\mathcal{A}) = \mathbb{E}_{p_x}[1_{\mathcal{A}}(x)],$$

where $1_{\mathcal{A}}$ is the indicator function of $\mathcal{A}$:

$$1_{\mathcal{A}}(x) = \begin{cases} 1, & x \in \mathcal{A}, \\ 0, & x \notin \mathcal{A}. \end{cases}$$

This identity states that the probability of $\mathcal{A}$ equals the expected value of its indicator function.

3.5 Expectation as an integral over the joint distribution

For any measurable function $g : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$,

$$\mathbb{E}_{p_{x,y}}[g(x, y)] = \int_{\mathcal{X}} \int_{\mathcal{Y}} g(x, y) p_{x,y}(x, y) dy dx,$$

where $p_{x,y}(x, y)$ is the joint density of x and y .

3.6 Integral of an indicator function times a conditional density

Let x and y be random variables taking values in $\mathcal{X}$ and $\mathcal{Y}$, respectively. For any set-valued event $\mathcal{A}(x) \subseteq \mathcal{Y}$ that may depend on x , define

$$1_{\mathcal{A}(x)}(y) = \begin{cases} 1, & y \in \mathcal{A}(x), \\ 0, & y \notin \mathcal{A}(x). \end{cases}$$

Then

$$\int_{\mathcal{Y}} 1_{\mathcal{A}(x)}(y) p_{y|x}(y | x) dy = \int_{\{y: y \in \mathcal{A}(x)\}} p_{y|x}(y | x) dy = \mathbb{P}(y \in \mathcal{A}(x) | x = x).$$

Equivalently,

$$\int_{\mathcal{A}(x)} p_{y|x}(y | x) dy = \mathbb{P}(y \in \mathcal{A}(x) | x = x).$$

3.7 Law of total probability for expectations

For any measurable function $g : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$,

$$\mathbb{E}_{p_{x,y}}[g(x, y)] = \int_{\mathcal{X}} \mathbb{E}_{p_{y|x=x}}[g(x, y)] p_x(x) dx,$$

where

$$\mathbb{E}_{p_{y|x=x}}[g(x, y)] = \int_{\mathcal{Y}} g(x, y) p_{y|x}(y | x) dy.$$

This expresses the overall expectation as an integral over realized values x of the conditional expectation given $x = x$, weighted by the marginal density $p_x(x)$.

3.8 Law of total expectation

If x and y are random variables on the same probability space, then

$$\mathbb{E}_{p_x}[x] = \mathbb{E}_{p_y}[\mathbb{E}_{p_{x|y}}[x | y]].$$

3.9 Conditional expectation

For random variables $x \in \mathcal{X}$ and $y \in \mathcal{Y}$,

$$\mathbb{E}_{p_{y|x=x}}[y] = \int_{\mathcal{Y}} y p_{y|x}(y | x) dy.$$

The conditional expectation of y given $x = x$ is the expected value of y under the conditional density $p_{y|x}(\cdot | x)$.

More generally, the conditional expectation as a deterministic function is

$$g(x) = \mathbb{E}_{p_{y|x=x}}[y],$$

and the conditional expectation given the random variable x is the random quantity

$$g(x) = \mathbb{E}_{p_{y|x}}[y | x].$$

3.10 Variance decomposition

For deterministic functions f and f_p ,

$$\begin{aligned} \mathbb{E}_{p_{x,y}}[(f(x) - y)^2] &= \mathbb{E}_{p_{x,y}}[(f(x) - f_p(x))^2] + \mathbb{E}_{p_{x,y}}[(f_p(x) - y)^2] \\ &\quad + 2\mathbb{E}_{p_{x,y}}[(f(x) - f_p(x))(f_p(x) - y)]. \end{aligned}$$

If the cross term satisfies

$$\mathbb{E}_{p_{x,y}}[(f(x) - f_p(x))(f_p(x) - y)] = 0,$$

then

$$\mathbb{E}_{p_{x,y}}[(f(x) - y)^2] = \mathbb{E}_{p_{x,y}}[(f(x) - f_p(x))^2] + \mathbb{E}_{p_{x,y}}[(f_p(x) - y)^2].$$

This decomposes total expected squared error into the error due to using f instead of f_p , and the irreducible error in predicting y from x .

3.11 Law of total variance

If x and y are random variables on the same probability space, and y has finite variance, then

$$\text{Var}_{p_y}(y) = \mathbb{E}_{p_x}[\text{Var}_{p_{y|x}}(y | x)] + \text{Var}_{p_x}(\mathbb{E}_{p_{y|x}}[y | x]).$$

Saying that x and y are on the same probability space means that both are functions of the same underlying random outcome $\omega \in \Omega$. It does not mean that they take values in the same kind of set. One random variable may be discrete and another may be continuous. What matters is that a joint law for (x, y) has been specified.

3.12 Example where the joint law is already specified

Let x be a coin flip taking values H and T, with

$$\mathbb{P}(x = H) = p, \quad \mathbb{P}(x = T) = 1 - p.$$

Given $x = H$, let

$$y | x = H \sim \mathcal{N}(\mu_h, \sigma_h^2),$$

and given $x = T$, let

$$y | x = T \sim \mathcal{N}(\mu_t, \sigma_t^2).$$

Although x is discrete and y is continuous, they are on the same probability space because the model describes one experiment: first draw the coin flip x , then draw y from the conditional distribution determined by the realized coin flip. Equivalently, the conditional laws define the

joint law

$$p_{x,y}(\mathbf{H}, y) := p \varphi_{\mu_h, \sigma_h^2}(y), \quad p_{x,y}(\mathbf{T}, y) := (1-p) \varphi_{\mu_t, \sigma_t^2}(y),$$

where φ_{μ, σ^2} is the density of $\mathcal{N}(\mu, \sigma^2)$. The resulting object is a joint distribution for the pair (x, y) on $\{\mathbf{H}, \mathbf{T}\} \times \mathbb{R}$.

For the first term in the law of total variance,

$$\text{Var}_{p_{y|x=\mathbf{H}}}(y) = \sigma_h^2, \quad \text{Var}_{p_{y|x=\mathbf{T}}}(y) = \sigma_t^2.$$

Therefore

$$\mathbb{E}_{p_x} \left[\text{Var}_{p_{y|x}}(y | x) \right] = p\sigma_h^2 + (1-p)\sigma_t^2.$$

For the second term,

$$\mathbb{E}_{p_{y|x=\mathbf{H}}}[y] = \mu_h, \quad \mathbb{E}_{p_{y|x=\mathbf{T}}}[y] = \mu_t.$$

Thus the random quantity $\mathbb{E}_{p_{y|x}}[y | x]$ takes value μ_h with probability p and value μ_t with probability $1-p$. Hence

$$\begin{aligned} \text{Var}_{p_x} \left(\mathbb{E}_{p_{y|x}}[y | x] \right) &= p(\mu_h - (p\mu_h + (1-p)\mu_t))^2 + (1-p)(\mu_t - (p\mu_h + (1-p)\mu_t))^2, \\ &= p(1-p)(\mu_h - \mu_t)^2. \end{aligned}$$

Therefore the law of total variance gives

$$\text{Var}_{p_y}(y) = p\sigma_h^2 + (1-p)\sigma_t^2 + p(1-p)(\mu_h - \mu_t)^2.$$

3.13 When a coupling is needed

A coupling is needed when x and y are originally specified separately, with no joint law between them. For example, suppose we are only given $x \sim \text{Bernoulli}(p)$ on one probability space, and separately $y \sim \mathcal{N}(0, 1)$ on another probability space. At this point, expressions such as

$$\mathbb{P}(y \leq 0 | x = 1)$$

are not yet defined, because no joint experiment has been specified.

A coupling creates new random variables with the same marginal distributions, but now defined on one common probability space. We use tildes to emphasize that these are new coupled copies:

$$\tilde{x} \text{ has the same distribution as } x, \quad \tilde{y} \text{ has the same distribution as } y.$$

The point is that $(\tilde{x}, \tilde{y})$ has a joint distribution, whereas the original separately specified x and y did not. One possible coupling is the independent coupling. Define the new sample space

$$\tilde{\Omega} := \{0, 1\} \times \mathbb{R}.$$

A point in this space is a pair (x, y) , where x is the Bernoulli outcome and y is the Gaussian outcome. Define the probability law by

$$p_{\tilde{x}, \tilde{y}}(x, y) := p_x(x)p_y(y).$$

On this new probability space, define

$$\tilde{x}(x, y) := x, \quad \tilde{y}(x, y) := y.$$

Thus $\tilde{x}$ reads off the first coordinate and $\tilde{y}$ reads off the second coordinate. The resulting coupled pair satisfies

$$\tilde{x} \sim \text{Bernoulli}(p), \quad \tilde{y} \sim \mathcal{N}(0, 1), \quad \tilde{x} \perp\!\!\!\perp \tilde{y}.$$

Now conditional probabilities involving both variables are well-defined. For any event $\mathcal{E} \subseteq \mathbb{R}$,

$$\mathbb{P}(\tilde{y} \in \mathcal{E}) = \mathbb{P}(\tilde{y} \in \mathcal{E} \mid \tilde{x} = 1)\mathbb{P}(\tilde{x} = 1) + \mathbb{P}(\tilde{y} \in \mathcal{E} \mid \tilde{x} = 0)\mathbb{P}(\tilde{x} = 0).$$

For the independent coupling, we can further reduce this using

$$\mathbb{P}(\tilde{y} \in \mathcal{E} \mid \tilde{x} = 1) = \mathbb{P}(\tilde{y} \in \mathcal{E} \mid \tilde{x} = 0) = \mathbb{P}(\tilde{y} \in \mathcal{E})$$

The tildes are only bookkeeping: they remind us that we have replaced separately defined random variables by coupled copies on one shared probability space. Different couplings can give different joint laws.

3.14 Definition of conditional probability

For events $\mathcal{A}, \mathcal{B}$, then

$$\mathbb{P}(\mathcal{A} \cap \mathcal{B}) = \mathbb{P}(\mathcal{A} \mid \mathcal{B})\mathbb{P}(\mathcal{B}).$$

3.15 Chain rule for factoring probability intersection

For events $\mathcal{A}_1, \dots, \mathcal{A}_n$, the definition of conditional probability can be extended to get the chain rule

$$\mathbb{P}(\mathcal{A}_1 \cap \mathcal{A}_2 \cap \dots \cap \mathcal{A}_n) = \mathbb{P}(\mathcal{A}_1)\mathbb{P}(\mathcal{A}_2 \mid \mathcal{A}_1)\mathbb{P}(\mathcal{A}_3 \mid \mathcal{A}_1 \cap \mathcal{A}_2) \dots \mathbb{P}(\mathcal{A}_n \mid \mathcal{A}_1 \cap \dots \cap \mathcal{A}_{n-1}),$$

whenever all conditional probabilities are well-defined. For example,

$$\mathbb{P}(\mathcal{A} \cap \mathcal{B} \cap \mathcal{C}) = \mathbb{P}(\mathcal{A})\mathbb{P}(\mathcal{B} \mid \mathcal{A})\mathbb{P}(\mathcal{C} \mid \mathcal{A} \cap \mathcal{B}).$$

The same intersection can be expanded in any ordering, as long as each later term conditions on all events already placed before it.

3.16 Chain rule for factoring joint densities

For two random variables x and y , the joint density can be factored as

$$p_{x,y}(x, y) = p_x(x) p_{y|x}(y \mid x).$$

For three random variables, the same rule gives

$$p_{x,y,z}(x, y, z) = p_x(x) p_{y|x}(y \mid x) p_{z|x,y}(z \mid x, y).$$

More generally, for random variables $x_1, \dots, x_n$, the joint density can be factored as

$$p_{x_1, \dots, x_n}(x_1, \dots, x_n) = p_{x_1}(x_1) p_{x_2|x_1}(x_2 \mid x_1) p_{x_3|x_1, x_2}(x_3 \mid x_1, x_2) \dots p_{x_n|x_1, \dots, x_{n-1}}(x_n \mid x_1, \dots, x_{n-1}).$$

3.17 Law of total probability

Let $\mathcal{A}_1, \dots, \mathcal{A}_k$ be disjoint events that partition the sample space Ω :

$$\mathcal{A}_1 \cup \dots \cup \mathcal{A}_k = \Omega, \quad \mathcal{A}_i \cap \mathcal{A}_j = \emptyset \quad \text{for } i \neq j.$$

Then for any event $\mathcal{B}$, the law of probability states that

$$\mathbb{P}(\mathcal{B}) = \sum_{j=1}^k \mathbb{P}(\mathcal{B} \mid \mathcal{A}_j) \mathbb{P}(\mathcal{A}_j).$$

This decomposes the probability of $\mathcal{B}$ by conditioning on which case $\mathcal{A}_j$ occurs.

3.18 Bayes' rule

Bayes' rule follows by writing $\mathbb{P}(\mathcal{A} \cap \mathcal{B})$ in two ways:

$$\mathbb{P}(\mathcal{A} \mid \mathcal{B}) \mathbb{P}(\mathcal{B}) = \mathbb{P}(\mathcal{B} \mid \mathcal{A}) \mathbb{P}(\mathcal{A}).$$

In general, if $\mathcal{A}_1, \dots, \mathcal{A}_k$ form a partition of Ω , then

$$\mathbb{P}(\mathcal{A}_j \mid \mathcal{B}) = \frac{\mathbb{P}(\mathcal{B} \mid \mathcal{A}_j) \mathbb{P}(\mathcal{A}_j)}{\sum_{\ell=1}^k \mathbb{P}(\mathcal{B} \mid \mathcal{A}_\ell) \mathbb{P}(\mathcal{A}_\ell)}.$$

Here $\mathbb{P}(\mathcal{A}_j)$ is the prior probability of hypothesis $\mathcal{A}_j$, and $\mathbb{P}(\mathcal{A}_j \mid \mathcal{B})$ is the posterior probability after observing $\mathcal{B}$.

Conditioning on another random variable w gives the conditional version:

$$p_{x,y|w}(x, y \mid w) = p_{x|w}(x \mid w) p_{y|x,w}(y \mid x, w).$$

Equivalently,

$$p_{x,y|w}(x, y \mid w) = p_{y|w}(y \mid w) p_{x|y,w}(x \mid y, w).$$

3.19 Add-subtract expectation trick

If a, b, c are random variables, then

$$\begin{aligned} \mathbb{E}_{p_{a,b,c}}[(a - b)^2] &= \mathbb{E}_{p_{a,b,c}}[(a - c)^2] + \mathbb{E}_{p_{a,b,c}}[(c - b)^2] \\ &\quad + 2 \mathbb{E}_{p_{a,b,c}}[(a - c)(c - b)]. \end{aligned}$$

When the cross term vanishes,

$$\mathbb{E}_{p_{a,b,c}}[(a - b)^2] = \mathbb{E}_{p_{a,b,c}}[(a - c)^2] + \mathbb{E}_{p_{a,b,c}}[(c - b)^2].$$

The cross term vanishes in common cases:

- Constant-times-zero-mean: if one factor is deterministic and the other has mean zero.
- Independence and zero mean: if $(a - c)$ and $(c - b)$ are independent and at least one has mean zero.

3.20 Hoeffding's inequality

Let $x_1, \dots, x_n$ be independent random variables with $x_i \in [a_i, b_i]$ almost surely, and let

$$s_n = \sum_{i=1}^n (x_i - \mathbb{E}_{p_{x_i}}[x_i]).$$

Then for all $t \geq 0$, Hoeffding's inequality says

$$\mathbb{P}(s_n \geq t) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right).$$

1. We can use the union bound to control the event $|s_n| \geq t$ by multiplying the one-sided bound by 2.
2. Hoeffding gives a distribution-free bound: only the ranges $b_i - a_i$ matter.
3. The probability behaves like a sub-Gaussian tail because it has the form $\exp(-ct^2)$.
4. Hoeffding ignores the actual variance; if the x_i have much smaller variance than their ranges suggest, Hoeffding can be loose.

3.21 Bernstein's inequality

Let $x_1, \dots, x_n$ be independent, mean-zero random variables with $|x_i| \leq M$ almost surely, and define

$$V = \sum_{i=1}^n \text{Var}_{p_{x_i}}(x_i).$$

Then for all $t \geq 0$, Bernstein's inequality says

$$\mathbb{P}\left(\sum_{i=1}^n x_i \geq t\right) \leq \exp\left(-\frac{t^2}{2\left(V + \frac{Mt}{3}\right)}\right).$$

1. We can use the union bound to control the event $|\sum_{i=1}^n x_i| \geq t$ by multiplying the one-sided bound by 2.
2. It interpolates between sub-Gaussian and sub-exponential tails:

$$t \ll \frac{V}{M} \quad \Rightarrow \quad \exp\left(-\frac{t^2}{2V}\right), \quad t \gg \frac{V}{M} \quad \Rightarrow \quad \exp\left(-\frac{3t}{2M}\right).$$

3. It is variance-adaptive: when V is small relative to M^2n , it can be much sharper than Hoeffding.
4. Inverting the one-sided bound, using $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$, and applying the union bound gives that, with probability at least $1 - \delta$,

$$\sum_{i=1}^n x_i \lesssim \sqrt{2V \log(1/\delta)} + \frac{2M}{3} \log(1/\delta).$$

3.22 Pushforward of a distribution

Let $x \sim p_x$ be a random variable and let

$$y = T(x)$$

for some deterministic map T . The distribution of y is called the pushforward of p_x through T , written

$$p_y = T_{\#}p_x.$$

This means that p_y is the distribution obtained by sampling $x \sim p_x$ and then applying T .

Equivalently, for any measurable set $\mathcal{A}$,

$$p_y(\mathcal{A}) = \mathbb{P}(y \in \mathcal{A}) = \mathbb{P}(T(x) \in \mathcal{A}) = p_x(T^{-1}(\mathcal{A})).$$

In particular, if $p_x(x)$ is the density, $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is invertible, and T is differentiable, then $p_y = T_{\#}p_x$ has density

$$p_y(y) = p_x(T^{-1}(y)) \left| \det \left(\frac{\partial}{\partial y} T^{-1}(y) \right) \right|.$$

For example, let $x \sim \text{Uniform}([0, 1])$, so

$$p_x(x) = 1_{[0,1]}(x).$$

Now define

$$T(x) = 2x + 1.$$

Then $y = T(x) = 2x + 1$ takes values in $[1, 3]$. Since

$$T^{-1}(y) = \frac{y-1}{2}, \quad \frac{d}{dy} T^{-1}(y) = \frac{1}{2},$$

the density of y is

$$p_y(y) = p_x\left(\frac{y-1}{2}\right) \frac{1}{2}.$$

Since $(y-1)/2 \in [0, 1]$ exactly when $y \in [1, 3]$, we get

$$p_y(y) = \frac{1}{2} 1_{[1,3]}(y).$$

Thus $T_{\#}p_x$ is the uniform distribution on $[1, 3]$.

3.23 Convolution with a Gaussian

A common way to smooth a distribution is to add a small amount of Gaussian noise. Let

$$y = x + z, \quad x \sim p_x, \quad z \sim \mathcal{N}(0, \sigma^2 I_d),$$

where x and z are independent. If φ_σ denotes the density of $\mathcal{N}(0, \sigma^2 I_d)$, then the density of y is

$$p_y(y) = (p_x * \varphi_\sigma)(y) = \int_{\mathbb{R}^d} p_x(x) \varphi_\sigma(y - x) dx.$$

This formula says that y can land near y if $x = x$ and the Gaussian noise moves x to y . Think of $p_y(y)$ as a local average of p_x around y .

A simple one-dimensional example is a point mass at zero:

$$x \sim \delta_0, \quad z \sim \mathcal{N}(0, \sigma^2), \quad y = x + z.$$

Since $x = 0$ with probability one, we have $y = z$, and therefore

$$p_y = \delta_0 * \varphi_\sigma = \varphi_\sigma.$$

So convolving a point mass with a Gaussian turns the point mass into a Gaussian bump.

4 Properties of random variables

4.1 Moment generating functions

The moment generating function of a random variable x is

$$M_x(t) := \mathbb{E}_{p_x} \left[e^{tx} \right].$$

It is useful because derivatives of M_x at $t = 0$ recover the moments of x :

$$M_x(0) = 1, \quad M_x^{(k)}(0) = \mathbb{E}_{p_x} [x^k].$$

For deterministic constants $a, b \in \mathbb{R}$,

$$M_{ax+b}(t) = e^{bt} M_x(at).$$

If x and y are independent, then the MGF of their sum factors:

$$M_{x+y}(t) = M_x(t) M_y(t).$$

4.2 Quantile function

Let

$$F_x(x) = \mathbb{P}(x \leq x)$$

be the cumulative distribution function of x . Then the quantile function is defined as

$$Q_x(p) := \inf \{x \in \mathbb{R} : F_x(x) \geq p\}, \quad p \in (0, 1).$$

If F_x is continuous and strictly increasing, then

$$Q_x(p) = F_x^{-1}(p)$$

on $(0, 1)$. Recall that the CDF is always increasing, but not always strictly increasing.

Essential identities for a general CDF:

- $F_x(Q_x(p)) \geq p$ and $Q_x(F_x(x)) \leq x$.
- Q_x is nondecreasing and left-continuous on $(0, 1)$.

When x has a density p_x with $p_x(Q_x(p)) > 0$, then

$$Q'_x(p) = \frac{1}{p_x(Q_x(p))} \quad \text{a.e. on } (0, 1).$$

Transformations:

- Inverse-transform sampling: if $u \sim \text{Unif}(0, 1)$, then

$$Q_x(u) \sim p_x.$$

- Location-scale equivariance: for $y = a + bx$,

$$Q_y(p) = \begin{cases} a + bQ_x(p), & b > 0, \\ a + bQ_x(1 - p), & b < 0. \end{cases}$$

Median: A median is any m with

$$F_x(m^-) \leq \frac{1}{2} \leq F_x(m),$$

since we might not have an m such that $F_x(m) = 1/2$. Equivalently, a median is any m such that

$$\mathbb{P}(x \geq m) \geq \frac{1}{2} \quad \text{and} \quad \mathbb{P}(x \leq m) \geq \frac{1}{2}.$$

This does not require the distribution to be continuous. If we do have continuity, then we can upgrade to equality: a median is any $m \in \mathbb{R}$ such that

$$\mathbb{P}(x \leq m) = \int_{-\infty}^m p_x(x) dx = \frac{1}{2}, \quad \mathbb{P}(x \geq m) = \int_m^{\infty} p_x(x) dx = \frac{1}{2}.$$

Named quantiles: Lower and upper quartiles are

$$Q_x\left(\frac{1}{4}\right), \quad Q_x\left(\frac{3}{4}\right).$$

For example, $Q_x(1/4)$ is the smallest value m such that a random draw from the distribution has probability at least $1/4$ of being at most m .

Empirical quantile for data $x^{(1)}, \dots, x^{(n)}$:

$$\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n 1\{x^{(i)} \leq x\}, \quad \hat{Q}_n(p) = \inf\{x : \hat{F}_n(x) \geq p\}.$$

4.3 Probability generating function trick

Let x be a nonnegative integer-valued random variable with probability generating function

$$G_x(t) = \mathbb{E}_{p_x}[t^x].$$

Then

$$\begin{aligned} \mathbb{E}_{p_x} \left[\frac{1}{x+1} \right] &= \sum_{k \geq 0} \frac{1}{k+1} \mathbb{P}(x = k) \\ &= \sum_{k \geq 0} \mathbb{P}(x = k) \int_0^1 t^k dt \\ &= \int_0^1 \mathbb{E}_{p_x}[t^x] dt \\ &= \int_0^1 G_x(t) dt. \end{aligned}$$

The intuition is to replace the reciprocal (before taking expectation) using

$$\frac{1}{k+1} = \int_0^1 t^k dt,$$

which turns the sum into an integral of the probability generating function.

Quick examples:

$$\begin{aligned}x \sim \text{Pois}(\lambda) & \Rightarrow \mathbb{E}_{p_x} \left[\frac{1}{x+1} \right] = \frac{1 - e^{-\lambda}}{\lambda}, \\x \sim \text{Bin}(n, p) & \Rightarrow \mathbb{E}_{p_x} \left[\frac{1}{x+1} \right] = \frac{1 - (1-p)^{n+1}}{p(n+1)}, \\x \sim \text{Geom}(p) \quad \text{failures before first success} & \Rightarrow \mathbb{E}_{p_x} \left[\frac{1}{x+1} \right] = \frac{p}{1-p} \log \frac{1}{p}.\end{aligned}$$

Useful bound from Jensen's inequality: Since $f(x) = 1/(1+x)$ is convex on $(-1, \infty)$,

$$\mathbb{E}_{p_x} \left[\frac{1}{1+x} \right] \geq \frac{1}{1 + \mathbb{E}_{p_x}[x]}.$$

5 Variance, covariance, correlation, and related topics

5.1 Independence of random variables

Let x and y be random variables on the same probability space. We say x and y are independent, written

$$x \perp\!\!\!\perp y,$$

if for every event $\mathcal{E}_x$ determined by x and every event $\mathcal{E}_y$ determined by y ,

$$\mathbb{P}(\mathcal{E}_x \cap \mathcal{E}_y) = \mathbb{P}(\mathcal{E}_x)\mathbb{P}(\mathcal{E}_y).$$

A useful consequence of independence is that,

$$\mathbb{E}_{p_{x,y}}[xy] - \mathbb{E}_{p_x}[x] \mathbb{E}_{p_y}[y] = 0$$

The reverse implication is not true in general!

5.2 Covariance

For two random variables x and y , the covariance is

$$\text{Cov}_{p_{x,y}}(x, y) = \mathbb{E}_{p_{x,y}}[(x - \mu_x)(y - \mu_y)].$$

Equivalently,

$$\text{Cov}_{p_{x,y}}(x, y) = \mathbb{E}_{p_{x,y}}[xy] - \mathbb{E}_{p_x}[x]\mathbb{E}_{p_y}[y].$$

We say x and y are uncorrelated if

$$\text{Cov}_{p_{x,y}}(x, y) = 0.$$

In general,

$$\text{Cov}_{p_{x,y}}(x, y) \in (-\infty, \infty),$$

but given variances, Cauchy–Schwarz gives

$$|\text{Cov}_{p_{x,y}}(x, y)| \leq \sqrt{\text{Var}_{p_x}(x) \text{Var}_{p_y}(y)}.$$

5.3 Correlation

Let x and y be random variables. Their correlation is

$$\text{Corr}_{p_{x,y}}(x, y) := \frac{\text{Cov}_{p_{x,y}}(x, y)}{\sqrt{\text{Var}_{p_x}(x) \text{Var}_{p_y}(y)}} \in [-1, 1].$$

Equivalently, correlation is the covariance of the standardized variables:

$$\text{Corr}_{p_{x,y}}(x, y) = \text{Cov}_{p_{x,y}}\left(\frac{x - \mathbb{E}_{p_x}[x]}{\sqrt{\text{Var}_{p_x}(x)}}, \frac{y - \mathbb{E}_{p_y}[y]}{\sqrt{\text{Var}_{p_y}(y)}}\right).$$

Thus correlation is a scale-free measure of linear association: it does not change if we rescale or shift either variable. Also, $|\text{Corr}_{p_{x,y}}(x, y)| = 1$ when y is an affine function of x . Correlation is dimensionless.

5.4 Independence $\implies$ zero covariance $\iff$ zero correlation

Assuming finite second moments and positive variances,

$$x \perp\!\!\!\perp y \quad \Rightarrow \quad \text{Cov}_{p_{x,y}}(x, y) = 0 \quad \iff \quad \text{Corr}_{p_{x,y}}(x, y) = 0.$$

Does zero covariance imply independence? In general, zero covariance does not imply independence:

$$\text{Cov}_{p_{x,y}}(x, y) = 0 \not\Rightarrow x \perp\!\!\!\perp y.$$

Zero covariance only rules out linear association; dependence can remain through nonlinear relationships.

Standard counterexample: Let $x \sim \text{Unif}(-1, 1)$ and define

$$y = x^2.$$

Then y is a deterministic function of x , so x and y are not independent. However,

$$\text{Cov}_{p_{x,y}}(x, y) = \mathbb{E}_{p_{x,y}}[xy] - \mathbb{E}_{p_x}[x]\mathbb{E}_{p_y}[y] = \mathbb{E}_{p_x}[x^3] - \mathbb{E}_{p_x}[x]\mathbb{E}_{p_x}[x^2] = 0.$$

This holds because

$$\mathbb{E}_{p_x}[x] = 0 \quad \text{and} \quad \mathbb{E}_{p_x}[x^3] = 0$$

by symmetry of the uniform distribution on $(-1, 1)$.

When does zero covariance imply independence? It does hold under additional structure, for example if (x, y) is jointly Gaussian:

$$(x, y) \text{ jointly normal} \quad \text{and} \quad \text{Cov}_{p_{x,y}}(x, y) = 0 \quad \Rightarrow \quad x \perp\!\!\!\perp y.$$

5.5 Confounding variables

Let (x, y, z) be random variables defined on a common probability space, where

1. x is a proposed cause,
2. y is an outcome,
3. z is a confounder, typically unobserved or uncontrolled.

Intuitively, z influences both x and y , creating an association between x and y that need not reflect a direct causal effect of x on y . A precise way to formalize this is via the law of total covariance:

$$\text{Cov}_{p_{x,y}}(x, y) = \mathbb{E}_{p_z} \left[\text{Cov}_{p_{x,y|z}}(x, y | z) \right] + \text{Cov}_{p_z} \left(\mathbb{E}_{p_{x|z}}[x | z], \mathbb{E}_{p_{y|z}}[y | z] \right).$$

Here $\text{Cov}_{p_{x,y|z}}(x, y | z)$ is a random variable, since it is a function of z , defined by

$$\text{Cov}_{p_{x,y|z}}(x, y | z) := \mathbb{E}_{p_{x,y|z}} \left[\left(x - \mathbb{E}_{p_{x|z}}[x | z] \right) \left(y - \mathbb{E}_{p_{y|z}}[y | z] \right) \mid z \right].$$

Interpretation of confounding variables:

1. $\mathbb{E}_{p_z}[\text{Cov}_{p_{x,y|z}}(x, y | z)]$ measures the within- z association: the average association between x and y after conditioning on the confounder z .

2. $\text{Cov}_{p_z}(\mathbb{E}_{p_{x|z}}[x | z], \mathbb{E}_{p_{y|z}}[y | z])$ measures the between- z association: the component of covariance induced solely by how variation in z shifts the conditional means of x and y .

A confounder corresponds precisely to the second term being nonzero. In particular, it is possible that

$$\text{Cov}_{p_{x,y|z}}(x, y | z) = 0 \quad \text{a.s.}$$

while

$$\text{Cov}_{p_z}(\mathbb{E}_{p_{x|z}}[x | z], \mathbb{E}_{p_{y|z}}[y | z]) \neq 0,$$

so that

$$\text{Cov}_{p_{x,y}}(x, y) \neq 0$$

entirely due to the influence of z on both variables.

5.6 Covariance measures strength of linear relationship

A quick way to see why covariance captures the strength of a linear relationship is to view it through least squares. For random variables x and y , consider approximating y by an affine function of x ,

$$\hat{y} = a + bx,$$

and choose (a, b) to minimize

$$\mathbb{E}_{p_{x,y}}[(y - a - bx)^2].$$

Differentiating in a gives

$$a = \mathbb{E}_{p_y}[y] - b \mathbb{E}_{p_x}[x].$$

Plugging this back in, the optimal slope is

$$b^* = \frac{\text{Cov}_{p_{x,y}}(x, y)}{\text{Var}_{p_x}(x)}, \quad \text{assuming } \text{Var}_{p_x}(x) > 0.$$

Thus covariance is exactly the numerator that determines the best linear slope relating y to x . Since covariance depends on scale, it is common to normalize it to obtain the correlation.

5.7 Empirical covariance formula

Let

$$x_1, \dots, x_n \in \mathbb{R}^d$$

be i.i.d. random vectors $x \sim p_x$ with mean

$$\mu_x = \mathbb{E}_{p_x}[x]$$

and covariance

$$\Sigma = \mathbb{E}_{p_x}[(x - \mu_x)(x - \mu_x)^\top].$$

Define the sample mean and the unbiased sample covariance by

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \quad s_{\text{unb}} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x})^\top.$$

We compute $\mathbb{E}_{p_{x_1, \dots, x_n}}[\mathbf{s}_{\text{unb}}]$:

$$\begin{aligned}\mathbb{E}_{p_{x_1, \dots, x_n}}[\mathbf{s}_{\text{unb}}] &= \mathbb{E}_{p_{x_1, \dots, x_n}} \left[\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x})^\top \right] \\ &= \frac{1}{n-1} \mathbb{E}_{p_{x_1, \dots, x_n}} \left[\sum_{i=1}^n ((x_i - \mu_x) - (\bar{x} - \mu_x)) ((x_i - \mu_x) - (\bar{x} - \mu_x))^\top \right].\end{aligned}$$

Expanding the square, and using

$$\sum_{i=1}^n (x_i - \mu_x) = n(\bar{x} - \mu_x),$$

we get

$$\begin{aligned}\mathbb{E}_{p_{x_1, \dots, x_n}}[\mathbf{s}_{\text{unb}}] &= \frac{1}{n-1} \mathbb{E}_{p_{x_1, \dots, x_n}} \left[\sum_{i=1}^n (x_i - \mu_x)(x_i - \mu_x)^\top - n(\bar{x} - \mu_x)(\bar{x} - \mu_x)^\top \right] \\ &= \frac{1}{n-1} \left(n\Sigma - n \cdot \frac{\Sigma}{n} \right) \\ &= \frac{1}{n-1} (n-1)\Sigma \\ &= \Sigma.\end{aligned}$$

where line 2 uses

$$\begin{aligned}\mathbb{E}_{p_{x_i}}[(x_i - \mu_x)(x_i - \mu_x)^\top] &= \Sigma, \\ \mathbb{E}_{p_{x_1, \dots, x_n}}[(\bar{x} - \mu_x)(\bar{x} - \mu_x)^\top] &= \text{Cov}_{p_{x_1, \dots, x_n}}(\bar{x}) = \frac{\Sigma}{n}.\end{aligned}$$

A similar calculation using the biased estimator

$$\mathbf{s}_{\text{bias}} = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x})^\top$$

gives

$$\mathbb{E}_{p_{x_1, \dots, x_n}}[\mathbf{s}_{\text{bias}}] = \frac{n-1}{n} \Sigma.$$

Thus the biased estimator underestimates the covariance by a constant factor in all feature dimensions.

6 Conjugate Priors

A prior family is *conjugate* to a likelihood family if multiplying the prior by the likelihood gives a posterior in the same distributional family as the prior. The basic pattern is

$$p_{\theta}(\theta) p_{\mathbf{x}|\theta}(\mathbf{x} | \theta) \propto p_{\theta|\mathbf{x}}(\theta | \mathbf{x}),$$

and conjugacy means

$$\text{prior} \times \text{likelihood} = \text{posterior} \in \text{same family as the prior.}$$

6.1 The exponential-family template

A useful way to recognize conjugacy is through the exponential-family form. First choose a prior with the exponential form

$$p_{\theta}(\theta) \propto \exp(\eta(\theta)^{\top} \nu - \tau A(\theta)),$$

for hyperparameters ν, τ . Then suppose the likelihood for one observation is

$$p_{\mathbf{x}|\theta}(x | \theta) = h(x) \exp(\eta(\theta)^{\top} T(x) - A(\theta)),$$

where, crucially, there is the same dependence on $\eta(\theta)$, $A(\theta)$. Then for an i.i.d. vector of data $\mathbf{x} = (x_1, \dots, x_n)$, the likelihood is

$$p_{\mathbf{x}|\theta}(\mathbf{x} | \theta) \propto \exp\left(\eta(\theta)^{\top} \sum_{i=1}^n T(x_i) - nA(\theta)\right).$$

Multiplying prior and likelihood gives

$$p_{\theta}(\theta) p_{\mathbf{x}|\theta}(\mathbf{x} | \theta) \propto \exp\left(\eta(\theta)^{\top} \left(\nu + \sum_{i=1}^n T(x_i)\right) - (\tau + n)A(\theta)\right).$$

Thus the posterior is in the same family as the prior, with updated hyperparameters

$$\nu_n = \nu + \sum_{i=1}^n T(x_i), \quad \tau_n = \tau + n.$$

6.2 Beta–Bernoulli

Take the Beta prior

$$\theta \sim \text{Beta}(\alpha, \beta), \quad p_{\theta}(\theta) \propto \theta^{\alpha-1} (1-\theta)^{\beta-1}, \quad \theta \in (0, 1).$$

Suppose the Bernoulli likelihood is

$$x_i | \theta \sim \text{Bernoulli}(\theta).$$

For a vector of i.i.d. data $\mathbf{x} = (x_1, \dots, x_n)$ with $s = \sum_{i=1}^n x_i$, the likelihood is

$$p_{\mathbf{x}|\theta}(\mathbf{x} | \theta) \propto \theta^s (1-\theta)^{n-s}.$$

Multiplying prior and likelihood gives the posterior

$$p_{\theta}(\theta) p_{\mathbf{x}|\theta}(\mathbf{x} | \theta) \propto \theta^{\alpha+s-1} (1-\theta)^{\beta+n-s-1}.$$

Therefore the posterior is

$$\theta | \mathbf{x} \sim \text{Beta}(\alpha + s, \beta + n - s),$$

which is in the same Beta family as the prior. Some useful summaries are

$$\mathbb{E}_{p_{\theta}}[\theta] = \frac{\alpha}{\alpha + \beta}, \quad \mathbb{E}_{p_{\theta|\mathbf{x}}}[\theta] = \frac{\alpha + s}{\alpha + \beta + n}.$$

6.3 Gamma–Poisson

Take the Gamma prior

$$\theta \sim \text{Gamma}(\alpha, \beta), \quad p_{\theta}(\theta) \propto \theta^{\alpha-1} e^{-\beta\theta}, \quad \theta > 0,$$

using the rate parameterization. Suppose the Poisson likelihood is

$$\mathbf{x}_i | \theta \sim \text{Poisson}(\theta).$$

For a vector of i.i.d. data $\mathbf{x} = (x_1, \dots, x_n)$ with $s = \sum_{i=1}^n x_i$, the likelihood is

$$p_{\mathbf{x}|\theta}(\mathbf{x} | \theta) \propto \theta^s e^{-n\theta}.$$

Multiplying prior and likelihood gives the posterior

$$p_{\theta}(\theta) p_{\mathbf{x}|\theta}(\mathbf{x} | \theta) \propto \theta^{\alpha+s-1} e^{-(\beta+n)\theta}.$$

Therefore the posterior is

$$\theta | \mathbf{x} \sim \text{Gamma}(\alpha + s, \beta + n),$$

which is in the same Gamma family as the prior. The posterior mean is

$$\mathbb{E}_{p_{\theta|\mathbf{x}}}[\theta] = \frac{\alpha + s}{\beta + n}.$$

6.4 Gamma–Exponential

Take the Gamma prior

$$\theta \sim \text{Gamma}(\alpha, \beta), \quad p_{\theta}(\theta) \propto \theta^{\alpha-1} e^{-\beta\theta}, \quad \theta > 0.$$

Suppose the Exponential likelihood with rate θ is

$$\mathbf{x}_i | \theta \sim \text{Exponential}(\theta).$$

For a vector of i.i.d. data $\mathbf{x} = (x_1, \dots, x_n)$, the likelihood is

$$p_{\mathbf{x}|\theta}(\mathbf{x} | \theta) \propto \theta^n e^{-\theta \sum_{i=1}^n x_i}.$$

Multiplying prior and likelihood gives the posterior

$$p_{\theta}(\theta) p_{\mathbf{x}|\theta}(\mathbf{x} | \theta) \propto \theta^{\alpha+n-1} e^{-(\beta+\sum_{i=1}^n x_i)\theta}.$$

Therefore the posterior is

$$\theta \mid \mathbf{x} \sim \text{Gamma}\left(\alpha + n, \beta + \sum_{i=1}^n x_i\right),$$

which is in the same Gamma family as the prior.

6.5 Gamma–Normal precision

Take the Gamma prior

$$\theta \sim \text{Gamma}(\alpha, \beta), \quad p_\theta(\theta) \propto \theta^{\alpha-1} e^{-\beta\theta}, \quad \theta > 0.$$

Suppose the Normal likelihood with known mean μ and unknown precision θ is

$$x_i \mid \theta \sim \mathcal{N}(\mu, \theta^{-1}).$$

For a vector of i.i.d. data $\mathbf{x} = (x_1, \dots, x_n)$, the likelihood is

$$p_{\mathbf{x}|\theta}(\mathbf{x} \mid \theta) \propto \theta^{n/2} \exp\left(-\frac{\theta}{2} \sum_{i=1}^n (x_i - \mu)^2\right).$$

Multiplying prior and likelihood gives the posterior

$$p_\theta(\theta) p_{\mathbf{x}|\theta}(\mathbf{x} \mid \theta) \propto \theta^{\alpha+n/2-1} \exp\left(-\left(\beta + \frac{1}{2} \sum_{i=1}^n (x_i - \mu)^2\right) \theta\right).$$

Therefore the posterior is

$$\theta \mid \mathbf{x} \sim \text{Gamma}\left(\alpha + \frac{n}{2}, \beta + \frac{1}{2} \sum_{i=1}^n (x_i - \mu)^2\right),$$

which is in the same Gamma family as the prior.

6.6 Normal–Normal with known variance

Take the Normal prior

$$\theta \sim \mathcal{N}(\mu_0, \tau_0^2).$$

Suppose the Normal likelihood with known variance σ^2 is

$$x_i \mid \theta \sim \mathcal{N}(\theta, \sigma^2).$$

For a vector of i.i.d. data $\mathbf{x} = (x_1, \dots, x_n)$, the likelihood is

$$p_{\mathbf{x}|\theta}(\mathbf{x} \mid \theta) \propto \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \theta)^2\right).$$

Therefore the posterior is

$$\theta \mid \mathbf{x} \sim \mathcal{N}(\mu_n, \tau_n^2),$$

which is in the same Normal family as the prior, where

$$\frac{1}{\tau_n^2} = \frac{1}{\tau_0^2} + \frac{n}{\sigma^2}, \quad \mu_n = \tau_n^2 \left(\frac{\mu_0}{\tau_0^2} + \frac{n\bar{x}}{\sigma^2} \right).$$

6.7 Normal–Inverse-Gamma for unknown mean and variance

Take the Normal–Inverse-Gamma prior

$$\sigma^2 \sim \text{Inv-Gamma}(\alpha_0, \beta_0), \quad \mu \mid \sigma^2 \sim \mathcal{N}\left(\mu_0, \frac{\sigma^2}{\kappa_0}\right).$$

Suppose the Normal likelihood with unknown mean and variance is

$$x_i \mid \mu, \sigma^2 \sim \mathcal{N}(\mu, \sigma^2).$$

For a vector of i.i.d. data $\mathbf{x} = (x_1, \dots, x_n)$, the likelihood is

$$p_{\mathbf{x} \mid \mu, \sigma^2}(\mathbf{x} \mid \mu, \sigma^2) \propto (\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2\right).$$

Therefore the posterior is Normal–Inverse-Gamma:

$$\sigma^2 \mid \mathbf{x} \sim \text{Inv-Gamma}(\alpha_n, \beta_n), \quad \mu \mid \sigma^2, \mathbf{x} \sim \mathcal{N}\left(\mu_n, \frac{\sigma^2}{\kappa_n}\right),$$

which is in the same Normal–Inverse-Gamma family as the prior, where

$$\kappa_n = \kappa_0 + n, \quad \mu_n = \frac{\kappa_0 \mu_0 + n \bar{x}}{\kappa_0 + n}, \quad \alpha_n = \alpha_0 + \frac{n}{2},$$

and

$$\beta_n = \beta_0 + \frac{1}{2} \sum_{i=1}^n (x_i - \bar{x})^2 + \frac{\kappa_0 n}{2(\kappa_0 + n)} (\bar{x} - \mu_0)^2.$$

6.8 Dirichlet–Categorical and Dirichlet–Multinomial

Take the Dirichlet prior

$$\boldsymbol{\theta} = (\theta_1, \dots, \theta_K) \sim \text{Dirichlet}(\alpha_1, \dots, \alpha_K),$$

with density

$$p_{\boldsymbol{\theta}}(\boldsymbol{\theta}) \propto \prod_{k=1}^K \theta_k^{\alpha_k - 1}, \quad \sum_{k=1}^K \theta_k = 1.$$

Suppose the Categorical likelihood is

$$\mathbb{P}(x = k \mid \boldsymbol{\theta}) = \theta_k, \quad k = 1, \dots, K.$$

For a vector of i.i.d. data $\mathbf{x} = (x_1, \dots, x_n)$ with counts $n_1, \dots, n_K$, the likelihood is

$$p_{\mathbf{x} \mid \boldsymbol{\theta}}(\mathbf{x} \mid \boldsymbol{\theta}) \propto \prod_{k=1}^K \theta_k^{n_k}.$$

Multiplying prior and likelihood gives the posterior

$$p_{\boldsymbol{\theta}}(\boldsymbol{\theta}) p_{\mathbf{x} \mid \boldsymbol{\theta}}(\mathbf{x} \mid \boldsymbol{\theta}) \propto \prod_{k=1}^K \theta_k^{\alpha_k + n_k - 1}.$$

Therefore the posterior is

$$\boldsymbol{\theta} \mid x \sim \text{Dirichlet}(\alpha_1 + n_1, \dots, \alpha_K + n_K),$$

which is in the same Dirichlet family as the prior. The posterior mean is

$$\mathbb{E}[\theta_k \mid x] = \frac{\alpha_k + n_k}{\sum_{j=1}^K \alpha_j + n}.$$